# Global Adaptive Second-Order Transformer for Remote Sensing Image Semantic Segmentation

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***Abstract*— In the domain of remote sensing (RS) image analysis, capturing global context is the key for precise semantic segmentation. Current vision transformer (ViT) advance this field by addressing convolutional neural network's (CNN) local receptive field limitations. However, ViT predominantly rely on the first-order information in image to establish global relationships, often overlooking the potential of second-order information, which is crucial for enhancing the discrimination of ground objects that exhibit high similarity and constant changes. To address this issue, we propose a global adaptive second-order transformer network (GASOT-Net). Specifically, the proposed global adaptive second-order transformer (GASOT) enhances the existing ViT structure by mining second-order information and adaptively fusing it with the first-order information during the process of establishing global dependency relationships. This approach enables the extraction of more discriminative features, thereby enriching the representation of global features. In addition, the local feature aggregation module (LFAM) is proposed to effectively aggregate features from different stages of CNN as input to the GASOT blocks. Moreover, to refine boundaries of complex ground objects, the global feature enhancement module (GFEM) is used in the decoder stage. In particular, GFEM includes two sub modules—feature shift module (FSM) and hierarchical feature fusion module (HFFM). FSM is used to enhance the local feature representation at first, and then, HFFM hierarchically aggregates local and global features from different stages. We conduct extensive experiments on four benchmark RS datasets, and the results show that our GASOT-Net outperforms other state-of-the-art methods. The code will be available at: https://github.com/j136812832/GASOT-Net.**

***Index Terms*— Feature enhancement, global feature, second-order transformer, semantic segmentation.**

## I. Introduction

With the development of remote sensing (RS) satellite technology, an increasing amount of RS image is being acquired. Therefore, there are increasingly more tasks involving the analysis and processing of RS images [1], [2], [3], [4], [5]. Semantic segmentation, as a fundamental computer vision task, plays a pivotal role in various fields including disaster management [6], [7], [8], environmental monitoring [9], [10], [11], [12], and urban planning [13], [14], [15]. Semantic segmentation of RS images is crucial for a more comprehensive understanding of the environment and land cover [16], [17], [18], [19], [20], [21], [22], [23].

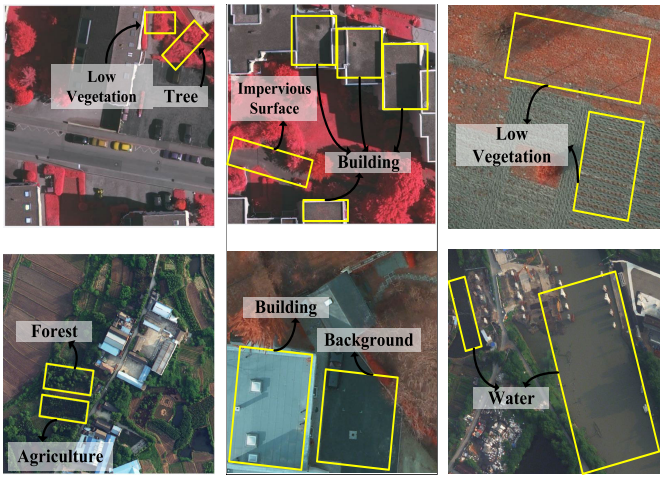



Fig. 1. Challenges faced by RS image semantic segmentation. The first row of images comes from the international society for photogrammetry and remote sensing (ISPRS) Vaihingen and Potsdam datasets, while the second row of images comes from the LoveDA dataset. (a) High similarity ground objects: low vegetation and tree, and forest and agriculture always appear simultaneously in a certain area. (b) High-similarity ground objects: building and impervious (background) have a highly similar appearance. (c) Constantly changing ground objects: the density of vegetation within the same area is inconsistent, and bodies of water are influenced by sunlight.

However, considering that ground objects often exhibit highly similar appearances or constantly change, the task of RS images semantic segmentation becomes notably challenging, as shown in Fig. 1. The global context information plays a crucial role in understanding the semantic relationships inherent in images, and previous studies have also demonstrated that capturing global context information from RS images can alleviate above challenges [24], [25].

In recent years, with the rapid advancement of deep learning [26], [27], [28], numerous methods have adopt convolutional neural network (CNN) architecture for semantic segmentation [29], [30], [31], [32], [33], [34], [35], [36]. Through multiple layers of convolution and downsample operations, CNN can progressively learn spatial position representation from

Manuscript received 8 July 2024; revised 4 August 2024; accepted 19 August 2024. Date of publication 2 September 2024; date of current version 26 September 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62071104, in part by the National Natural Science Foundation of China (NNSFC) and Civil Aviation Administration of China (CAAC) under Grant U2133211 and Grant U2233209, and in part by the Chengdu Key Research and Development Support Plan under Grant 2022-YF05-02120-SN. *(Corresponding author: Jian Cheng.)*

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Digital Object Identifier 10.1109/TGRS.2024.3453501

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images. However, CNN-based methods often face limitations in capturing global context information due to the restricted local receptive field [37], [38], [39].

The emergence of transformer [40] has partially addressed the issue of limited receptive fields in CNN. Initially, it was primarily used in tasks such as natural language processing and has been widely used in image processing in recent years [41], [42], [43], [44]. The vision transformer (ViT) [45] is a pioneering model that applies the transformer architecture to image processing. It divides the input image into fixed-sized patches and processes them using a series of self-attention layers, enabling effective feature extraction and achieving remarkable performance in image classification tasks. Subsequently, Zheng et al. [46] proposed segmentation transformer (SETR), which replaces the traditional CNN encoders with pure transformer-structured encoders, thereby altering the existing semantic segmentation architecture. Compared with the previous semantic segmentation models, SETR demonstrates superior contextual information capture ability, further enhancing the accuracy and efficiency of semantic segmentation. Swin Transformer [47] is another transformer-based architecture that introduces a hierarchical structure with shifted windows to capture spatial information effectively. It employs self-attention mechanisms across different scales to capture both local and global contextual information for accurate segmentation. The success of transformer in the field of computer vision has sparked many scholars to apply it in the field of RS image processing [25], [48], [49], [50]. However, the majority of current transformers primarily focus on the first-order information, failing to establish deeper global dependencies within images. This limitation always hinders their effectiveness in segmenting ground objects that exhibit high similarity and constant changes, due to insufficient discriminative information in global feature representation. Some scholars have confirmed that paying attention to second-order information in the process of establishing global relationships can obtain more discriminative features of the target [51], [52].

Based on the above assertions, this article proposes a global adaptive second-order transformer network (GASOT-Net). Specifically, the local feature aggregation module (LFAM) is used to effectively aggregate local features from different stages of ConvNeXt [53], [54], which can filter out redundant information and highlight important spatial information. The aggregated local features will enter continuous proposed several global adaptive second-order transformer (GASOT) blocks to capture a large amount of global contextual information. The GASOT has improved the existing ViT structure. On the one hand, it can mine second-order information during the establishment of global dependency relationships. By doing this, more discriminative information can be obtained to address the challenges posed by high similarity and constantly changing ground objects. On the other hand, the representation of global features is enriched by adaptively fusing second-order information and first-order information. Moreover, the global feature enhancement module (GFEM) is used to further enhance the representation of the extracted global features to refine boundary segmentation in the decoder stage. The GFEM includes two submodules—feature shift module (FSM) and hierarchical feature fusion module (HFFM). Specifically, FSM is used to expand the receptive field of local features from different stages at first, and then, HFFM is used to hierarchically aggregate the global and local features at different stages, which takes into account the information from different encoding layers to enhance spatial feature representation. Ultimately, the feature map will be restored to its original resolution to produce the final prediction result. The main contribution of this article is summarized as follows.

1) We propose an RS images semantic segmentation network that combines existing ConvNeXt and proposed GASOT, which can fully mine and utilize more enrichment and discriminative global context in RS images to achieve high-precision segmentation results.
2) We propose the GASOT, a novel transformer architecture tailored specifically to address the challenges arising from the high similarity and constant variation of ground objects in RS images. Moreover, it is adept at extracting more discriminative global features and adaptively combines first-order and second-order information in the process of establishing global relationships.
3) The LFAM, which we have designed as a novel approach, specializes in aggregating features from various stages of CNN. It can retain important spatial and local detail features while filtering out redundant features.
4) At the upsampling stage, we employ a novel GFEM that is specifically designed to obtain for fine boundary segmentation. Specifically, the FSM is used to strengthen local features by performing feature offset operations, and the HFFM is used to effectively aggregate global and local features from different stages.

## II. Related Works

### A. Semantic Segmentation With CNNs

The semantic segmentation has always been a highly focused task in the field of computer vision, providing powerful support for image analysis. Fully convolutional network (FCN) [29] is the earliest CNN network to achieve end-to-end semantic segmentation, and its emergence marks a new era in semantic segmentation. U-Net [31] compensates for the lost detail information during the encoding phase through skip connections. SegNet [30] is an encoder–decoder architecture for semantic segmentation that uses pooling indices from the encoder to improve upsampling efficiency in the decoder. In response to the phenomenon of insufficient receptive field, some methods use parallel computing to enhance the perception of global features, such as DeepLabV3+ [33] and PSPNet [55]. However, the above methods still cannot effectively obtain the dependency relationship between global and local features. Therefore, a large number of methods have begun to use attention mechanisms to establish global and local dependencies for global perception of images. SENet [56] calculates the weight coefficients corresponding to the channels of feature map through global adaptive pooling, thereby determining the dependency relationships between

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channels. DANet [57] constructs the long-range dependencies in images by using spatial attention and channel attention mechanisms. CCNet [35] introduces a criss-cross-attention module, enabling the capture of contextual information across the entire spatial dimensions of an image more efficiently and with fewer parameters, thus optimizing computational cost and performance. OCRNet [58] enhances feature representations for semantic segmentation by capturing object-contextual relationships, utilizing an object-contextual module to improve accuracy by focusing on both object and surrounding contextual information. Resolution is an essential factor in semantic segmentation of RS images. Inspired by HRNet [36], HighDAN [59] achieves cross-resolution information exchange and multiscale fusion by maintaining multiple resolution feature streams in parallel. It preserves the precise positional information of high-resolution feature maps and integrates the rich semantic information of low-resolution feature maps. In addition, many methods also focus on the research of real-time semantic segmentation. BiSeNetV2 [60] is a fast, accurate neural network for real-time semantic segmentation, balancing spatial details, and contextual information using dual paths. Xu et al. [61] proposed a real-time semantic segmentation network that can efficiently capture contextual, spatial, and boundary information in images.

Compared with natural images, semantic segmentation of RS images with many highly similar and constantly changing ground targets is more challenging. Su et al. [62] proposed a multifeature learning network for semantic segmentation of remote sensing (RS) images. This network leverages CNN to learn low-level semantic features, high-level semantic features, and discriminative features of ground targets. By doing this, it achieves accurate segmentation of similar ground objects. Zhao et al. [63] proposed a method that integrates multiple attention methods to explore the global and local dependency relationships in RS images. Li et al. [64] proposed a method that can gradually refine edge features in the process of multilevel feature aggregation, thereby obtaining segmentation results with complete boundaries.

In summary, CNN-based semantic segmentation methods have advantages in spatial position representation. Although many methods attempt to expand the receptive field of CNN to capture more global context information through attention mechanisms and multiscale parallel mechanisms, it is still far from enough for complex ground objects in RS images.

### *B. Vision Transformer*

Transformer was initially widely used in the field of natural language processing [40], [65], [66], which can effectively establish global dependencies between input and output sequences. Generally speaking, the transformer block is composed of multihead self-attention, a multilayer perceptron, and layer normalization.

In recent years, transformers have emerged as a powerful feature extractor for various computer vision tasks. One prominent application is image classification, where transformers have shown remarkable performance. ViT [45] replaces the traditional convolutional layers of CNN with transformer encoder, enabling the model to efficiently capture global image representations. It achieves impressive performance on large-scale image classification benchmarks, such as ImageNet. Another influential work is Swin Transformer [47], which leverages hierarchical architectures and window-based self-attention mechanisms to overcome the computational limitations of the original transformer in handling high-resolution images. It achieves state-of-the-art results on various image classification datasets. In the field of semantic segmentation, SETR [46] is a new type of semantic segmentation model based on ViT architecture, which uses a complete ViT structure as the backbone to achieve better contextual information capture and feature expression capabilities, providing new ideas for semantic segmentation tasks. Segformer [67] is a simple yet powerful semantic segmentation model that leverages transformers for efficient and accurate performance. It achieves real-time speeds while maintaining high accuracy. MaskFormer [68] is an advanced framework that revolutionizes semantic and instance segmentation tasks by leveraging mask classification. Its elegant approach simplifies complex segmentation challenges and achieves remarkable performance. Topformer [69] is a lightweight semantic segmentation method for mobile devices, which proposes the token pyramid module to capture local features, and the semantics extractor is used to model the global and local relationships of features. SegNeXt [70] is a recent state-of-the-art model that leverages the advancements in both CNN and transformers, combining their strengths to achieve superior segmentation performance. In the field of object detection, detection transformer (DETR) [71] is a method of end-to-end object detection using transformer. It transforms object detection into a sequence-to-sequence problem, using transformer encoders and decoders to achieve object detection and classification. DETR has achieved good performance in some designs of traditional object detection methods, such as avoiding anchor boxes and nonmaximum suppression. In recent years, many RS foundational models based on transformer structure have emerged. SpectralGPT [72] is a universal foundational model designed specifically for spectral RS images, utilizing an innovative 3-D generative pretrained transformer architecture. It is capable of processing spectral RS images of different sizes, resolutions, time series, and regions and has demonstrated significant performance improvements in downstream tasks.

In RS image processing, more and more methods are also using transformers to improve performance. ST-U-Net [25] combines Swin Transformer and U-Net to improve the feature discrimination of ground objects. Liu et al. [48] reconsidered the structure of transformers and proposed a method that can integrate global and local features in RS images. Mo et al. [73] combined Swin Transformer and DeepLabV3+, utilizing their respective advantages to capture multiscale features of terrain objects and construct global dependencies.

In summary, existing transformers excel at establishing long-range dependencies in images to capture abundant global features and have found extensive applications across various domains of computer vision. Nevertheless, they primarily focus on utilizing first-order information from images to establish global dependencies, thereby neglecting the significant

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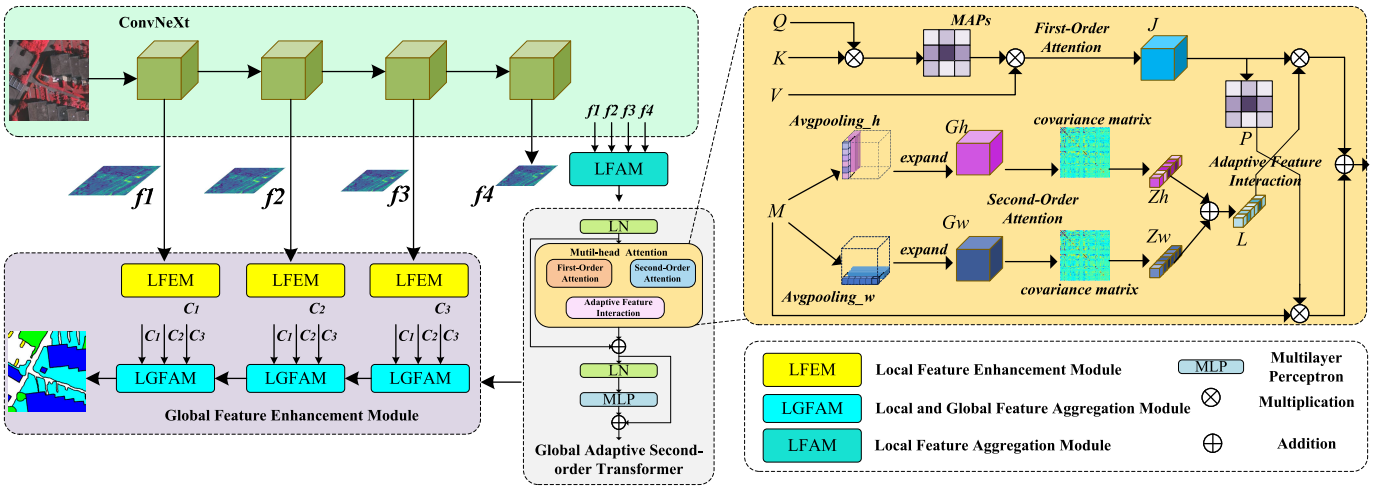



Fig. 2. GASOT-Net. The GASOT-Net uses ConvNeXt to capture local features of different stages in RS images. Subsequently, LFAM aggregates the local features from four stages. The GASOT enriches the representation of global features, while GFEM enhances the representation of these global features.

role of second-order information. Such an omission becomes particularly critical when attempting to accurately distinguish between ground objects that are highly similar in appearance.

### C. Second-Order Methods in Deep Learning

The information obtained by CNN through the continuous convolutional layers is often first-order information. Generally speaking, first-order information usually is obtained through convolutional operations, mainly focusing on local spatial information of the image. By contrast, the second-order information refers to more abstract and discriminative features with high-level representation capabilities.

Li et al. [51] proposed a second-order pooling method for classification tasks. Gao et al. [52] proposed a simple and effective pooling method—covariance pooling. Inserting this pooling in CNN can learn more discriminative presentations. He et al. [74] proposed a method called SCCov network for the RS image scene classification, which inserts the covariance pooling into the associated CNN to achieve learning of more representative features using high-order information. STLNet [75] uses the idea of histograms to rearrange image pixels and learns this arrangement using a CNN architecture to obtain rich texture features of the image.

In deep learning, it is crucial to comprehensively consider the dual effects of first-order and second-order information in order to obtain a more discriminative and accurate image representation. However, the existing transformer structures always overlook the mining of the second-order information.

## III. Methodology

### A. GASOT-Net

To address the issue of incorrect segmentation of ground objects with high similarity and constant changes in RS images, this article presents a novel encoder–decoder segmentation network that utilizes the proposed second-order transformer to comprehensively extract and integrate rich global features from RS images. Fig. 2 illustrates the overall structure of the proposed GASOT-Net.

We adopt ConvNeXT to extract local features $f_i (i = 1, 2, 3, 4)$ from the input image $I \in \mathbf{R}^{3\times H\times W}$, where $i$ represents different stages of CNN. In order to obtain more effective local feature representations, LFAM is used to aggregate local features from four stages from CNN. Then, the aggregated feature $F$ will enter $N$ consecutive second-order transformer blocks to achieve global feature mining. In every second-order transformer block, the size of the input and output features of remain consistent.

In the decoder stage, three successive GFEMs are employed to aggregate local features and global features. Specifically, each GFEM consists of an FSM and an HFFM. In FSM, local features extracted by the ConvNeXt are enhanced to obtain more robust local feature representations. Then, these enhanced features are fused with corresponding global features from the decoder stage in HFFM to obtain enhanced global feature representations. With each application of GFEM, the resolution of the feature map is doubled.

After undergoing three consecutive global feature enhancements, the final decoded result, denoted by $O \in \mathbf{R}^{3\times H/4\times W/4}$, is obtained. To maintain consistency with the original resolution, we perform a $4\times$ upsampling on $O$, resulting in a prediction result.

### B. ConvNeXt

ConvNeXT is a next-generation CNN backbone, which improves the structure of the previous CNN and achieves excellent performance in classification tasks. The comparison between the ConvNeXt block and ResNet block is shown in Fig. 3. ConvNeXt not only inherits the residual connection mechanism of ResNet to solve the gradient vanishing problem in deep network training but also makes significant improvements and innovations in multiple key aspects. First, ConvNeXt adjusted the proportion of blocks in each stage of the network structure to optimize the performance of the model. Second, ConvNeXt also introduces an inverse bottleneck layer structure, which can capture the feature information efficiently. In addition, ConvNeXt adopts large convolutional kernels and optimizes the microscopic design, such as using

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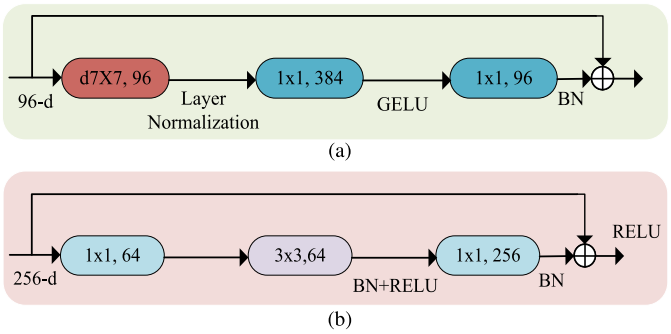



Fig. 3. Comparison of (a) ConvNeXt block and (b) ResNet block.

Gaussian error linear unit (GELU) activation functions and layer normalization, further improving the model's expressive power and computational efficiency.

The local features $f_i (i = 1, 2, 3,$ and $4)$ captured by ConvNeXt will enter the LFAM for feature aggregation on the one hand and enhance the perception ability of surrounding features through FSM on the other hand.

### C. Local Feature Aggregation Module

The encoding process in CNN is continuous, and the features of the next stage already include the features of the previous stage. However, some existing methods aggregate the features of different stages equally, which can lead to redundancy of information and neglect of critical information. Based on the above conclusions, we mainly focus on the local feature maps $f3$ and $f4$ and use a gate mechanism (sigmoid) to obtain the response degree of different features in $f1$ and $f2$, so that relevant features can be activated in $f3$ and $f4$. It is noteworthy that in our design, $f3$ will be activated by the weight of $f1$, and $f4$ will be activated by the weight of $f2$. Such a design is primarily based on two considerations. On the one hand, it is best not to have the feature maps of the interaction in two consecutive stages, as they will have similar features. On the other hand, it is also important not to have stages that are far apart, as the difference in features makes activation more difficult.

The structure of LFAM is shown in Fig. 4. First, we enhance the detailed features in $f1$ and $f2$ through a learnable Laplace filter with a learnable coefficient $\alpha$, which allows for a more flexible extraction of detailed features

$$f_i = f_i + (\alpha \cdot L) * f_i, \quad \text{for } i \in \{1, 2\} \tag{1}$$

where $L$ is the Laplace filter and $0 < \alpha < 1$ is the learnable coefficient.

Then, $f1$ and $f2$ execute the downsampling strategy, and the number of their channels becomes 1. After the above operation, the resolution of $f1$ remains consistent with $f3$, and the resolution of $f2$ remains consistent with $f4$. As shown in (2) and (3), our gate mechanism uses the sigmoid function to process $f1$ and $f2$ to obtain $v1$ and $v2$. The local feature maps $f3$ and $f4$ are multiplied by the processed maps of $v1$ and $v2$, respectively, to achieve filtering the redundant information and strengthening the useful spatial information. The results

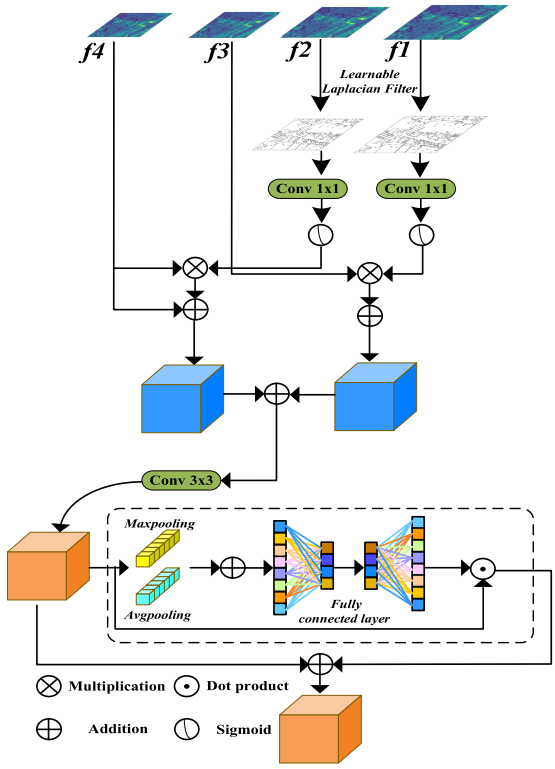



Fig. 4. Structure of LFAM.

of executing them separately will be added together to obtain the output $U$

$$v_i = \text{sigmoid}(f_i), \quad \text{for } i \in \{1, 2\} \tag{2}$$

$$U = [f_4 \oplus (v_2 * f_4)] \oplus [f_3 \oplus (v_1 * f_3)] \tag{3}$$

where $0 < v_i < 1$ represents the activation level of each pixel in the spatial dimension and $\oplus$ represents pixel-by-pixel addition.

Next, $U$ will enhance spatial detail information through global average pooling (GAP) and global max pooling (GMP), where GAP can obtain target level information, while GMP can obtain discriminative information of objects. The relevant operations are as follows:

$$U \odot [\text{FC}(\text{GAP}(U) \oplus \text{GMP}(U))] \tag{4}$$

where FC represents the fully connected layer and $\odot$ represents the dot product.

### D. GASOT Block

In our network, global features are extracted through a series of consecutively stacked GASOT blocks. Each GASOT block comprises a multihead attention module, a feed-forward network, and residual connections. Distinguishing itself from ViT, the multihead attention within the GASOT architecture incorporates a second-order attention mechanism, enabling the capture of more second-order information from images. Furthermore, GASOT uniquely possesses the capability to adaptively blend both first-order and second-order features, thus enriching the global feature representation.

Next, we will introduce our multihead attention module, which includes three parts—first-order attention, second-order

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attention, and adaptive feature interaction. We assume that the input is $I \in \mathbf{R}^{c\times h\times w}$. For the part of the first-order attention, we perform self-attention operations on it according to (5) and (6), where $\mathbf{T}$ is a learnable matrix and $\mathrm{MAP}_S$ is a spatial attention map. The output of the first-order attention $J$ can be obtained by (7), where $\otimes$ represents the matrix addition

$$[Q, K, V] = I\mathbf{T}_{QKV} \tag{5}$$

$$\mathrm{MAP}_S = \mathrm{softmax}\left(QK^{\top}/\sqrt{D_h}\right) \tag{6}$$

$$J = V \otimes \mathrm{MAP}_S. \tag{7}$$

For the part of the second-order attention, we first use $3 \times 3$ convolution on $I$ and obtain $M \in \mathbf{R}^{c\times h\times w}$. First, the intermediate feature $M$ collects spatial information in both vertical and horizontal directions by employing parallel GAP. The specific operations are depicted as follows:

$$g^z_{h_i} = \frac{1}{w}\sum_{j=0}^{w-1}\hat{M}^z(i, j) \tag{8}$$

$$g^z_{w_j} = \frac{1}{h}\sum_{i=0}^{h-1}\hat{M}^z(i, j), \quad 0 \le i < h,\ 0 \le j < h \tag{9}$$

where $i$ is the index of the vertical direction, $j$ is the index of the horizontal direction, and $z$ is the index of the channel. $\hat{M}$ is $f(M)$, where $f()$ represents the $3 \times 3$ convolution with batch normalization and the GELU activation function. Through the above calculation, $G_h \in \mathbf{R}^{c\times h\times 1}$ and $G_w \in \mathbf{R}^{c\times 1\times w}$ can be obtained.

Subsequently, the obtained results $G_h$ and $G_w$ are restored to the same size as $M$ through the expand operation. The expanded features $G_h$ and $G_w$ contain abundant spatial features, and it is necessary to delve deeper into the second-order information embedded within these global spatial information. We obtain second-order channel relationship weights for each feature map by calculating the covariance of $G_h$ and $G_w$ separately. Next, we use $G_h$ as an example to introduce the process of obtaining second-order channel relationship weights.

We reshape $G_h$ to obtain matrix $\mathbf{X} \in \mathbf{R}^{c\times(h\times w)}$. Then, we can obtain the covariance matrix $\mathbf{\Sigma}$ through the following calculation:

$$\mathbf{\Sigma} = \mathbf{X}\bar{\mathbf{I}}\mathbf{X}^{\mathrm{T}} \tag{10}$$

where $\mathbf{X}^{\mathrm{T}}$ is the transpose matrix of $\mathbf{X}$, and $\bar{\mathbf{I}} = (1/(h \times w))(\mathbf{I} - (1/(h \times w))\mathbf{1})$, $\mathbf{I}$ and $\mathbf{1}$ are the $(h \times w) \times (h \times w)$ identity matrix and matrix of all ones, respectively.

Then, we normalize the covariance matrix $\mathbf{\Sigma}$ to obtain more discriminative representations from the global features. Considering the covariance matrix $\mathbf{\Sigma}$ as symmetric and positive semidefinite, it undergoes the following eigenvalue decomposition (EIG):

$$\mathbf{\Sigma} = \mathbf{U}\mathbf{\Lambda}^{\alpha}\mathbf{U}^{\mathrm{T}} \tag{11}$$

where $\mathbf{U}$ represents an orthogonal matrix, while $\mathbf{\Lambda}$, defined as $\mathrm{diag}(\lambda_1, \ldots, \lambda_C)$, is a diagonal matrix comprising eigenvalues arranged in a descending order.

In addition, the covariance matrix can also be converted to the power of the eigenvalues as follows:

$$\hat{\mathbf{Y}} = \mathbf{\Sigma}^{\alpha} = \mathbf{U}\mathbf{\Lambda}^{\alpha}\mathbf{U}^{\mathrm{T}} = [\mathbf{y}_1, \ldots, \mathbf{y}_C] \tag{12}$$

where $\alpha$ is a positive real number. In this article, we set $\alpha = 1/2$. The channelwise statistics $Z_h \in \mathbf{R}^{C\times 1}$ can be obtained by shrinking $\hat{\mathbf{Y}}$, as follows:

$$Z_h = \frac{1}{C}\sum_i y_c(i). \tag{13}$$

The normalized covariance matrix can reflect the relationship between channels. Therefore, we use the normalized covariance matrix to construct the channel attention mechanism.

We employ a feature intersection mechanism to adaptively combine the complementary first-order and second-order information, thus obtaining a more comprehensive global feature representation. Specifically, $J$ and $Z$ are processed according to the following equations, respectively:

$$P = \mathrm{sigmoid}\,(\mathrm{conv}(J)) \tag{14}$$

$$L = \mathrm{sigmoid}\,(Z) \tag{15}$$

where $P \in \mathbf{R}^{1\times h\times w}$ and $L \in \mathbf{R}^{c\times 1\times 1}$, sigmoid represent the sigmoid function. conv represents $1 \times 1$ convolution, aimed at making the feature map channel count to 1.

Next, the calculation of the interaction between first-order and second-order features, and the final output is as follows:

$$O = \mathrm{conv}((J \odot L) \oplus (K \odot P)) \tag{16}$$

where $\odot$ represents the point multiplication, $\oplus$ represents the matrix addition, and $O$ represents the final result of the GASOT block.

### *E. Global Feature Enhancement Module*

GFEM includes two submodules, FSM and HFFM. In order to further enhance the representation of global features, we first use FSM to enhance the representation of local features, and then aggregate local and global features through HFFM to achieve global feature enhancement. The structure of GFEM is shown in Fig. 5.

*1) Feature Shift Module:* In order to further expand the receptive field of local features, FSM is used to enhance local features from different CNN stages. Specifically, we divide the local feature map into five groups along channel dimension, and then perform two parallel and independent feature shift operations. In one shift operation, the last four sets of features will move right, left, down, and up to expand the receptive field of local features, while the first set of features will not move. In another shift operation, the last four groups will move left, right, up, and down to further enhance the ability to perceive local features in opposite directions. The shift operation in each direction is achieved by setting the weight of a specific position in the convolutional kernel to 1, and the weight of other positions to 0. This setting allows the convolutional operation to only consider pixel values in specific directions in the input feature map, thereby achieving the shift effect of the feature map. The shifted features undergo $1 \times 1$ convolution and rectified linear unit (ReLU) activation function to obtain new features, and the two new sets of features are added together to obtain the final feature map. The above process

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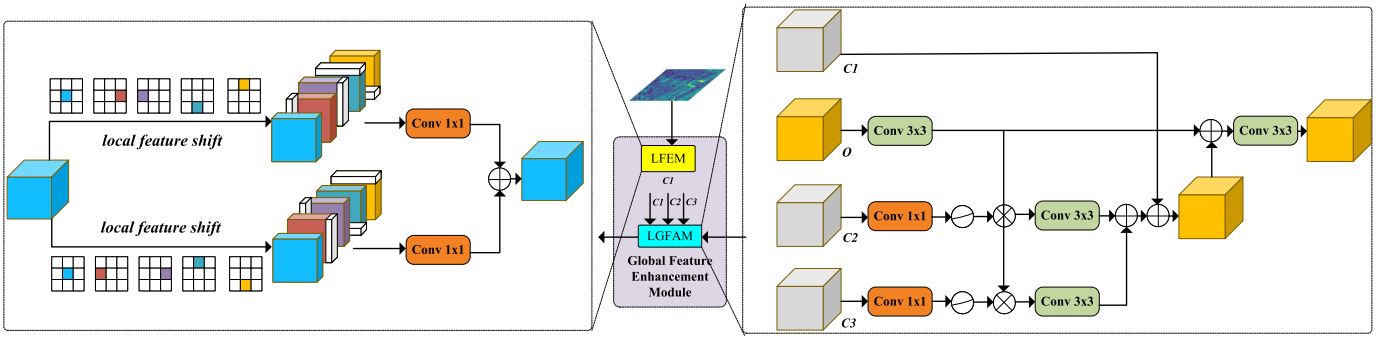



Fig. 5. GFEM.

description is as follows:

$$X = [X_1, X_2, X_3, X_4, X_5] \tag{17}$$

$$\begin{aligned} X_1^{(1)} &= X_1, \quad X_1^{(2)} = X_1 \\ X_2^{(1)} &= \text{shift_right}\,(X_2)\,, \quad X_2^{(2)} = \text{shift_left}\,(X_2) \\ X_3^{(1)} &= \text{shift_left}\,(X_3)\,, \quad X_3^{(2)} = \text{shift_right}\,(X_3) \\ X_4^{(1)} &= \text{shift_down}\,(X_4)\,, \quad X_4^{(2)} = \text{shift_up}\,(X_4) \\ X_5^{(1)} &= \text{shift_up}\,(X_5)\,, \quad X_5^{(2)} = \text{shift_down}\,(X_5) \end{aligned} \tag{18}$$

$$Y^{(1)} = \text{ReLU}\left(\text{Conv}_{1\times1}\left(\left[X_1^{(1)}, X_2^{(1)}, X_3^{(1)}, X_4^{(1)} X_5^{(1)}\right]\right)\right) \tag{19}$$

$$Y^{(2)} = \text{ReLU}\left(\text{Conv}_{1\times1}\left(\left[X_1^{(2)}, X_2^{(2)}, X_3^{(2)}, X_4^{(2)} X_5^{(2)}\right]\right)\right) \tag{20}$$

$$Y = Y^{(1)} + Y^{(2)} \tag{21}$$

where $X$ represents the input feature map and $Y$ represents the output feature map. By the feature-shifting operation, local features at different stages enhance the perception capabilities of surrounding features, which can strengthen their representation and lay a solid foundation for subsequent hierarchical feature fusion.

*2) Hierarchical Feature Fusion Module:* HFFM utilizes the same gate mechanism (sigmoid) as LFAM to filter local features from FSM. Unlike traditional encoder–decoder architectures with skip connections, HFFM considers not only the local features at the current stage but also those from other stages, thereby comprehensively enhancing spatial information. In other words, for the current decoding stage, a gate mechanism will be used to activate local features from the previous stages to preserve key semantic information.

It is important to note that the structure depicted in Fig. 5 represents the final decoder layer. Thus, the feature $C1$ will be added to $O$ from the preceding decoder layer, while features $C2$ and $C3$ will be activated via the gate mechanism. The above process description is as follows:

$$W_2 = \sigma(\text{Conv}_{1\times1}(C_2)) \tag{22}$$

$$W_3 = \sigma(\text{Conv}_{1\times1}(C_3)) \tag{23}$$

$$O' = \text{Conv}_{3\times3}(O) \tag{24}$$

$$F = \text{Conv}_{3\times3}(O' \odot W_2) + \text{Conv}_{3\times3}(O' \odot W_3) \tag{25}$$

$$K = F + C_1 \tag{26}$$

$$\text{Y} = \text{Conv}_{3\times3}(K + O') \tag{27}$$

where $\sigma$ represents the sigmoid, and $W_2$ and $W_3$ represent the weight activation map.

### F. Loss Function

We employ the online hard example mining cross-entropy loss (OhemCELoss) to optimize the network's learning process. Specifically, this loss function places greater emphasis on optimizing challenging samples during training by assigning higher training weights to them. The expression for OhemCELoss is as follows:

$$\mathcal{L}_{\text{OHEM}} = -\frac{1}{N}\sum_{i=1}^{N} \text{loss}_i \cdot \text{weight}_i \tag{28}$$

where loss represents the cross-entropy loss, $N$ represents the number of samples, and weight represents the weight of network.

## IV. Experiment

### A. Dataset

*1) Vaihingen Dataset:* The ISPRS Vaihingen dataset[1] contains a total of 33 aerial images of the city of Vaihingen an der Enz in Germany, with an average resolution of 2494 × 2064. It has a total of six annotated ground objects: impervious surfaces, buildings, low vegetation, trees, cars, and clutters/background. According to the dataset's official guidelines, the dataset is divided into a training set of 11 images and a test set of five images.

*2) Potsdam Dataset:* The ISPRS Potsdam dataset[2] is also a well-known high-resolution RS image annotation dataset released in 2018. This dataset collected aerial images of the city of Potsdam in Germany. The Potsdam dataset also categorizes the same six types of ground objects. The collected target range of this dataset is 38.4 km$^2$, with a spatial resolution of 0.5 m. In contrast to the Vaihingen dataset, the Potsdam dataset is more extensive, comprising 38 RS images each with dimensions of 6000 × 6000 pixels. According to the dataset's official partitioning, this dataset is segmented into 24 images for training and 14 for testing.

[1]https://www2.isprs.org/commissions/comm2/wg4/benchmark/2d-semlabel-vaihingen/

[2]2https://www2.isprs.org/commissions/comm2/wg4/benchmark/2d-semlabel-potsdam/

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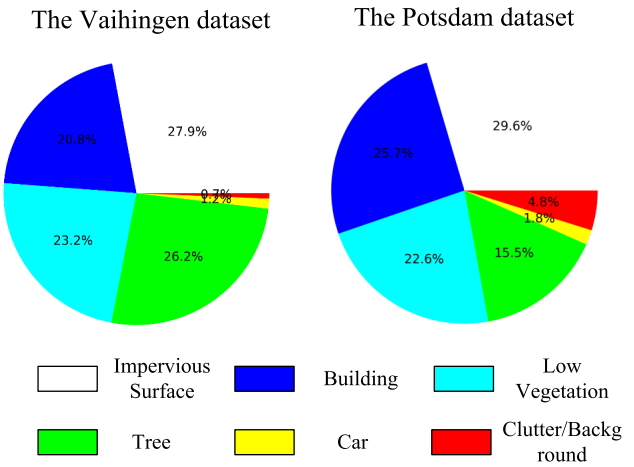



Fig. 6. Proportion of different ground objects on the ISPRS Vaihingen and Potsdam datasets.

*3) LoveDA Dataset:* The LoveDA dataset [76] is released in 2021 for RS image semantic segmentation, including 5987 0.3 m high-resolution images and 166 768 annotation semantic objects from three different cities in China (Nanjing, Guangzhou, and Wuhan). This dataset includes both urban and rural scenes, with complex ground objects and high-intraclass similarity. According to the dataset's official partitioning, we use 2552 images for training and 1669 images for validation.

*4) UAVid Dataset:* The UAVid dataset [77], tailored for urban semantic segmentation, offers a robust collection of high-resolution video footage captured by UAVs. Comprising 300 images with resolutions of 4096 × 2160 or 3840 × 2160 pixels, it targets eight distinct segmentation classes: buildings, roads, trees, low vegetation, static cars, moving cars, humans, and clutter. For training and validation, the official split provides 150 labeled training images and 50 labeled validation images. The remaining 100 labeled images constitute the undisclosed test set for benchmarking.

The ISPRS Vaihingen dataset and the Potsdam dataset have the same ground object labeling. However, the proportion of different ground objects varies in different datasets. Fig. 6 shows the respective proportions of different ground objects in the two datasets.

## B. Implementation Details

*1) Training Settings:* For the ISPRS Vaihingen dataset and Potsdam dataset, all images are cropped to a size of 512 × 512, maintaining a one-third overlap. For the LoveDA dataset, the size of the input image is already 512 × 512. The SGD optimizer is employed, with a weight decay parameter set at 0.0005. The batch size is set to 4 in all experiments. The number of GASOT blocks in our method is set to 4. We begin with an initial learning rate of 0.003. To preserve training stability and facilitate ongoing, effective model updates, we adopt the following strategy for decaying the learning rate:

$$\mathrm{lr} = \mathrm{lr}_{\mathrm{init}} \times \left(1 - \frac{\text{current_iterations}}{\text{max_iterations}}\right)^{0.9}. \tag{29}$$

*2) Evaluation Metrics:* To effectively evaluate the performance of different methods, we use three common evaluation metrics in semantic segmentation tasks. The evaluation metrics include mean $F1$ score, mean intersection over union (mIoU), and overall accuracy (OA), with their respective calculation methods detailed as follows:

$$\mathrm{OA} = \frac{N_{\mathrm{TP}} + N_{\mathrm{TN}}}{N_{\mathrm{TP}} + N_{\mathrm{FP}} + N_{\mathrm{FN}} + N_{\mathrm{TN}}} \tag{30}$$

$$\mathrm{IoU} = \frac{N_{\mathrm{TP}}}{N_{\mathrm{TP}} + N_{\mathrm{FP}} + N_{\mathrm{FN}}} \tag{31}$$

$$F_1 = \frac{2 * \left(\frac{N_{\mathrm{TP}}}{N_{\mathrm{TP}}+N_{\mathrm{FP}}} * \frac{N_{\mathrm{TP}}}{N_{\mathrm{TP}}+N_{\mathrm{FN}}}\right)}{\left(\frac{N_{\mathrm{TP}}}{N_{\mathrm{TP}}+N_{\mathrm{FP}}} + \frac{N_{\mathrm{TP}}}{N_{\mathrm{TP}}+N_{\mathrm{FN}}}\right)} \tag{32}$$

where $N_{\mathrm{TP}}$ represents the true tositive, $N_{\mathrm{FP}}$ represents the false positive, and $N_{\mathrm{FN}}$ represents the false negative.

$F1$ score is a metric that comprehensively considers both precision and recall and is very useful for evaluating the performance of semantic segmentation models. mIOU and OA can objectively measure the predictive accuracy of the model at the pixel level.

## C. Ablation Study

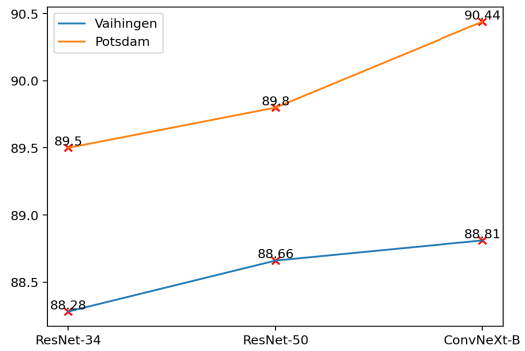
*1) Backbone:* Our proposed GASOT-Net uses CNN to extract local features from RS images. In many computer vision tasks, methods usually use ResNet as the backbone. ConvNeXt was proposed by Facebook AI in 2022, which rethought and improved the existing CNN structure to achieve higher classification accuracy on multiple datasets. We conduct comparative experiments on ResNet and ConvNeXt. The comparison networks we have chosen include ResNet-34, ResNet-50, and ConvNeXt-B. The experimental results are shown in Fig. 7.

Thanks to its superior network design, ConvNeXt performs better than ResNet in three datasets. Among them, ConvNeXt-B has the best value. In the subsequent experiments, our methods all use ConvNeXt-B to capture local features.

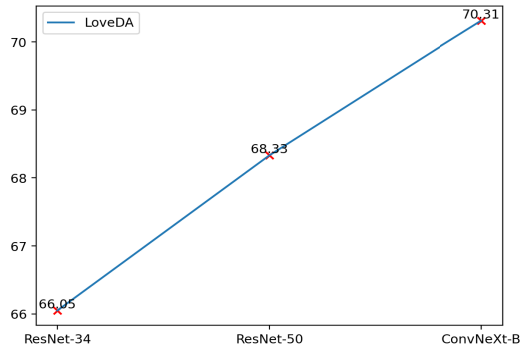
*2) Local Feature Aggregation Module:* LFAM is proposed to aggregate local features from different stages of CNN. To verify its effectiveness, we conduct comparative experiments. The experiments compared with LFAM included the following two groups. First, the resolution of $f1$–$f3$ is adjusted to be the same as $f4$ through downsampling, and then, the four feature maps are concatenated together. Finally, the channel of the result is adjusted through 1 × 1 convolution (Downsample + Concat + Conv). Second, the resolution of $f1$–$f3$ is also adjusted to be the same as $f4$ through downsampling, and then, convolution is used to adjust their channel count to be the same as $f4$. Finally, the four feature maps are added by pixel to obtain the final feature map (Downsample + Conv + Add).

The results of the experiment are shown in Table I. It can be clearly seen that using LFAM to aggregate local features of CNN is the best. This indicates that compared with simply adding or concating features, filtering out redundant features can better highlight the role of key features.

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(a)




(b)

Fig. 7. Comparative experiment of ConvNeXt and CNN. (a) Experiment is conducted on the ISPRS Vaihingen and Potsdam datasets. (b) Experiment is conducted on the LoveDA dataset.

TABLE I

COMPARATIVE EXPERIMENTS ON DIFFERENT AGGREGATION METHODS OF LOCAL FEATURES

| Method | Dataset | OA(%) | mIoU(%) | mean $F1$(%) |
|---|---|---|---|---|
| Downsample + Concat + Conv | Vaihingen | 89.10 | 74.68 | 84.65 |
| | Postdam | 90.49 | 85.96 | 90.92 |
| | LoveDA | 70.35 | 51.35 | 67.26 |
| Downsample + Conv+ Add | Vaihingen | 89.04 | 74.58 | 84.71 |
| | Postdam | 90.46 | 85.92 | 90.97 |
| | LoveDA | 70.31 | 51.47 | 67.15 |
| LFAM(Ours) | Vaihingen | 89.24 | 74.86 | 84.73 |
| | Postdam | 90.59 | 86.06 | 91.14 |
| | LoveDA | 70.48 | 51.67 | 67.50 |

*3) Global Adaptive Second-Order Transformer:* It can be clearly seen from Tables II–IV that with using the GASOT block, the network has a significant increase in OA, mIOU and $F1$ score. In addition, when using a GASOT, the performance of the network has also been improved compared with using ViT.

Fig. 8 shows the visualization results of the experiment. We have selected some representative samples to confirm the effectiveness of GASOT. In RS images, building and background (always road), low vegetation, and tree often have highly similar appearances. It is difficult for Baseline + ViT to accurately distinguish them based on local features and first-order information. Compared with Baseline + ViT, our method Baseline + GASOT is able to correctly segment these ground objects, thanks to the deep exploration of second-order information in the global feature extraction process, thus obtaining more distinguishing features about similar ground objects. In addition, some ground objects vary significantly in the same scene. Due to the injection of a large amount of second-order information at the global level, our method can also accurately understand the differences caused by these changes, thus achieving accurate segmentation. We highlighted the segmentation improvement in the above situation with a red bounding box in Fig. 8.

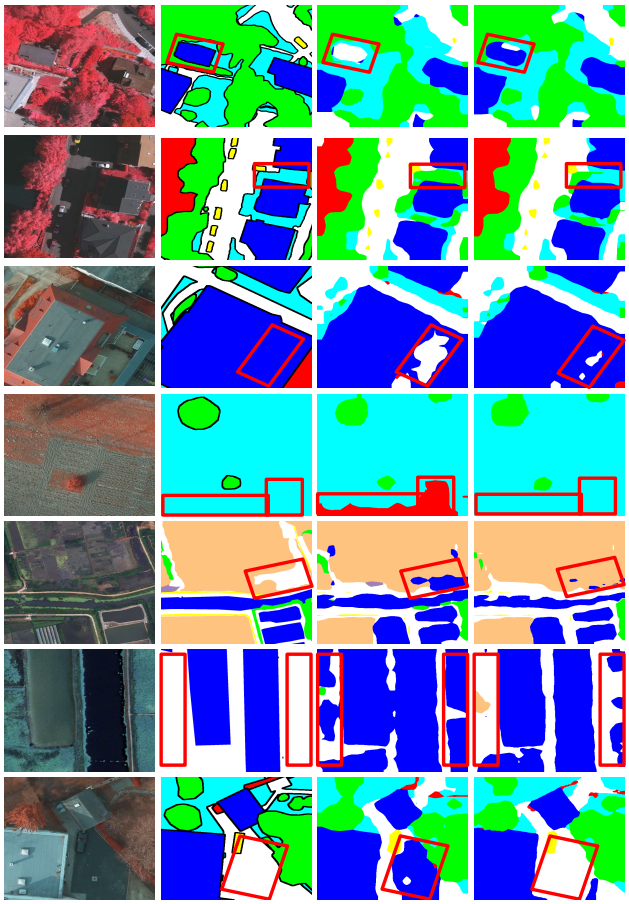


Fig. 8. Comparative experiment between ViT and GASOT. (Left–right) Input image, ground truth, Baseline + ViT, and Baseline + GASOT.

TABLE II

ABLATION STUDY OF SECOND-ORDER TRANSFORMER BLOCK ON THE VIHINGEN DATASET

| Method | OA(%) | mIOU(%) | mean $F1$(%) |
|---|---|---|---|
| Baseline | 88.80 | 74.24 | 84.02 |
| Baseline+ViT | 88.87 | 74.65 | 84.39 |
| Baseline+GASOT | **89.19** | **75.01** | **84.68** |

*4) Global Feature Enhancement Module:* We conducted ablation experiments on GFEM, including two submodules: FSM and HFFM. As shown in Tables V and VI, our baseline network does not perform well in OA, $F1$, and mIOU evaluation metrics. When we use FSM or HFFM, the effectiveness

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TABLE III
ABLATION STUDY OF SECOND-ORDER TRANSFORMER BLOCK ON THE POTSDAM DATASET

| Method | OA(%) | mIOU(%) | mean $F1$(%) |
|---|---|---|---|
| Baseline | 90.44 | 85.88 | 90.87 |
| Baseline+ViT | 90.57 | 86.07 | 91.03 |
| Baseline+GASOT | **90.88** | **86.32** | **91.28** |

TABLE IV
ABLATION STUDY OF SECOND-ORDER TRANSFORMER BLOCK ON THE LOVEDA DATASET

| Method | OA(%) | mIOU(%) | mean $F1$(%) |
|---|---|---|---|
| Baseline | 70.31 | 50.93 | 66.71 |
| Baseline+ViT | 71.87 | 52.60 | 68.18 |
| Baseline+GASOT | **72.29** | **54.03** | **69.49** |

TABLE V
ABLATION STUDY OF GFEM ON THE VAIHINGEN DATASET

| Method | OA(%) | mIOU(%) | mean $F1$(%) |
|---|---|---|---|
| Baseline | 88.80 | 74.24 | 84.02 |
| Baseline+LFEM | 90.03 | 81.02 | 88.88 |
| Baseline+LGFAM | 89.87 | 80.61 | 88.58 |
| Baseline+GFEM | **90.20** | **81.41** | **89.09** |

TABLE VI
ABLATION STUDY OF GFEM ON THE LOVEDA DATASET

| Method | OA(%) | mIOU(%) | mean $F1$(%) |
|---|---|---|---|
| Baseline | 70.31 | 50.93 | 66.71 |
| Baseline+FSM | 71.28 | 53.08 | 68.52 |
| Baseline+HFFM | 70.56 | 52.96 | 68.44 |
| Baseline+GFEM | **72.35** | **54.86** | **69.55** |

of the network is improved to varying degrees. Compared with other networks, our baseline with GFEM achieves the highest values in all three evaluation indicators. This indicates that the proposed GFEM can effectively aggregate local and global features from different stages, thereby achieving high results.

Fig. 9 shows the visualization of segmentation results. We can clearly see that when the GFEM module is removed, the segmentation of many ground objects is prone to incomplete segmentation and blurred boundaries. After using the GFEM module, it can be seen that the boundary segmentation of the ground object is clearer and more complete. This indicates that our proposed GFEM can effectively aggregate global and local features during the upsampling stage, ensuring the completeness of segmentation and clear boundary.

### D. Comparison With State-of-the-Art Methods

To verify the effectiveness of our proposed GASOT-Net, we conducted comparative experiments with other advanced methods on three RS semantic segmentation datasets. The methods that we choose for comparison can be divided into three groups: CNN-based methods, transformer-based methods, and the network for semantic segmentation of RS images. Specifically, CNN-methods include FCN, U-Net, DeepLabV3+, DANet, PSPNet, BiSeNet, and so on. Transformer methods include Segformer, SwinU-Net, Topformer, and so on. The network for semantic segmentation of RS images includes MFANet, HCANet, DLNet, and so on.

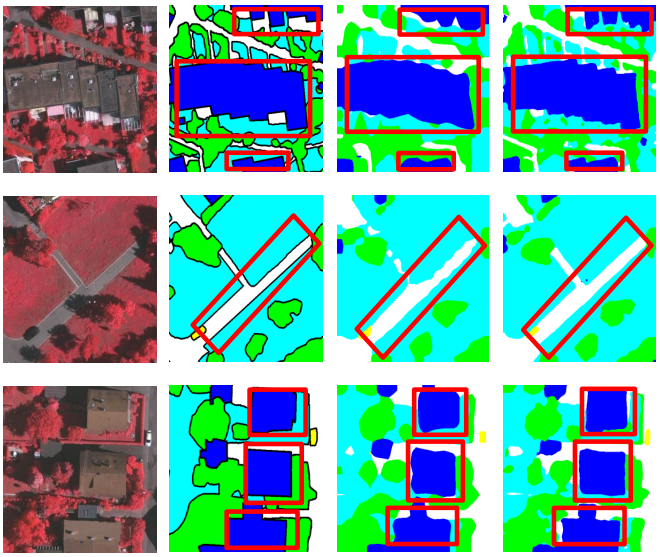



Fig. 9. Comparison of segmentation results of different models. (a) Image. (b) GT. (c) Baseline. (d) Baseline + GFEM.

In our comparative experiment, FCN, DeepLabV3+, PSPNet, DANet, and BiSeNet all used the ResNet-50 that was pretrained on imageNet as the backbone. Segformer, SwinU-Net, and Topforner use open-source pretrained weights.

Tables VII and VIII present the comparative experimental results on the Vaihingen and Potsdam datasets, respectively. Thanks to its powerful capability of global context information extraction, our proposed GASOT-Net has achieved the highest values in all indicators. Figs. 10 and 11, respectively, show the comparison of segmentation results of various models on the Vaihingen and Potsdam datasets. As basic segmentation models, the segmentation results of FCN and U-Net clearly have a lot of defects, such as blurred boundaries and incorrect classification of similar categories. DeepLabV3 and PSPNet have similar design concepts, using parallel dilated convolutions or pooling kernels of different sizes to expand the receptive field of the network, respectively. Benefiting from the expansion of the receptive field, DeepLabV3 and PSPNet have shown significant improvements. DANet uses an attention mechanism to construct global and local dependencies. Although the above methods have some improvements compared with FCN and U-Net, most of them are tailored for segmentation in natural scenes. RS images contain a large number of ground objects that exhibit high similarity and constant variation. When these methods are applied to RS images, their semantic segmentation performance is unsatisfactory. In contrast, our proposed GASOT-Net is capable of capturing high-order information from ground targets, enriching the representation of global features, and therefore achieving superior performance.

Compared with the ISPRS Vaihingen and Potsdam datasets, the LoveDA dataset is a more challenging RS image semantic

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TABLE VII
EXPERIMENTAL RESULTS ON THE VAIHINGEN DATASET

| Method | Impervious Surface | Building | Low Vegetation | Tree | Car | mean $F_1$(%) | mIoU(%) | OA(%) |
|---|---|---|---|---|---|---|---|---|
| FCN [29] | 90.21 | 94.08 | 81.15 | 87.34 | 62.40 | 83.04 | 73.41 | 88.30 |
| DLR [78] | 92.40 | 95.20 | 83.90 | 89.90 | 81.20 | 88.52 | - | 90.30 |
| UNet [31] | 90.06 | 92.05 | 81.50 | 87.68 | 74.07 | 85.07 | 75.69 | 87.79 |
| SegNet [30] | 86.08 | 91.52 | 77.94 | 88.61 | 76.00 | 84.03 | 74.17 | 86.98 |
| PSPNet [55] | 90.21 | 94.12 | 81.38 | 87.53 | 61.81 | 83.01 | 73.12 | 88.38 |
| DeeplabV3+ [33] | 90.33 | 94.39 | 81.56 | 87.53 | 63.20 | 83.40 | 73.54 | 88.48 |
| DANet [57] | 90.21 | 94.13 | 81.50 | 87.56 | 64.68 | 83.62 | 74.15 | 88.46 |
| BiSeNet [79] | 91.13 | 94.84 | 79.51 | 87.86 | 64.04 | 83.48 | 74.20 | 88.84 |
| HRNet [36] | 92.56 | 95.63 | 83.19 | 88.91 | 87.87 | 89.84 | 81.78 | 90.54 |
| Segformer-B5 [67] | 88.25 | 90.63 | 80.67 | 87.76 | 53.72 | 80.21 | 70.20 | 86.75 |
| Topformer [69] | 87.07 | 88.64 | 80.03 | 85.74 | 52.98 | 78.89 | 68.85 | 85.99 |
| TransUNet [37] | 86.99 | 89.21 | 79.91 | 87.05 | 58.31 | 80.29 | 68.69 | 85.74 |
| HCANet [80] | 92.47 | 95.01 | 84.21 | 89.37 | 84.02 | 89.02 | 81.03 | 90.33 |
| DLNet [82] | 91.29 | 95.25 | 78.78 | 87.91 | 79.68 | 86.58 | 76.99 | 88.71 |
| MFANet [82] | 92.72 | 95.77 | 84.23 | 89.27 | 84.68 | 89.34 | 81.64 | 90.60 |
| EMRT [1] | 89.16 | 92.68 | 78.69 | 86.54 | 80.12 | 85.44 | 74.96 | 86.97 |
| GASOT-Net(ConvNeXt-V1) | 93.41 | 95.99 | **85.16** | 89.80 | **89.12** | 90.70 | **83.36** | 91.02 |
| GASOT-Net(ConvNeXt-V2) | **93.58** | **96.18** | 85.06 | **89.86** | 89.10 | **90.76** | 83.30 | **91.35** |

TABLE VIII
EXPERIMENTAL RESULTS ON THE POTSDAM DATASET

| Method | Impervious Surface | Building | Low Vegetation | Tree | Car | mean $F_1$(%) | mIoU(%) | OA(%) |
|---|---|---|---|---|---|---|---|---|
| FCN [29] | 91.85 | 95.94 | 85.82 | 87.21 | 88.86 | 89.94 | 84.38 | 89.69 |
| V-FuseNet [82] | 92.70 | 96.30 | 87.30 | 88.50 | 95.40 | 92.04 | - | 90.60 |
| TSMTA [14] | 92.91 | 97.13 | 87.03 | 87.26 | 95.16 | 91.90 | - | 90.64 |
| TreeUNet [83] | 93.10 | 97.30 | 86.60 | 87.10 | 95.80 | 91.98 | - | 90.70 |
| CASIA3 [84] | 93.40 | 96.80 | 87.60 | 88.30 | 96.10 | 92.44 | - | 91.00 |
| UNet [31] | **93.58** | 96.77 | 86.87 | 88.59 | 96.24 | 84.36 | 74.95 | 88.46 |
| SegNet [30] | 91.62 | 94.71 | 84.69 | 86.98 | 94.46 | 90.49 | 82.88 | 88.60 |
| PSPNet [55] | 91.78 | 96.08 | 85.81 | 87.39 | 88.14 | 89.84 | 84.12 | 89.61 |
| DeeplabV3+ [33] | 92.95 | 95.88 | 87.62 | 88.15 | 96.02 | 92.12 | 84.32 | 90.88 |
| DANet [57] | 92.59 | 96.49 | 86.29 | 87.62 | 89.30 | 90.46 | 84.92 | 90.20 |
| HRNet [36] | 93.44 | 96.86 | 87.02 | 88.11 | 96.24 | 92.33 | 86.17 | 90.98 |
| BiSeNet [79] | 92.65 | 96.18 | 86.52 | 87.42 | 95.06 | 91.57 | 87.19 | 90.37 |
| Segformer-B5 [67] | 90.65 | 94.21 | 83.28 | 82.94 | 90.65 | 88.35 | 82.34 | 87.34 |
| TransUNet [37] | 90.17 | 93.35 | 83.31 | 83.68 | 91.61 | 88.42 | 79.52 | 87.10 |
| Topformer [69] | 90.03 | 93.78 | 82.95 | 82.45 | 90.06 | 87.85 | 82.01 | 87.12 |
| MFANet [81] | 93.24 | 96.50 | 87.05 | 88.19 | 95.16 | 92.03 | - | 90.88 |
| EMRT [1] | 90.87 | 94.86 | 84.12 | 85.23 | 90.32 | 89.08 | 80.54 | 88.12 |
| GASOT-Net(ConvNeXt-V1) | 93.44 | 97.14 | **87.99** | 88.96 | 96.35 | 92.78 | 89.05 | 91.39 |
| GASOT-Net(ConvNeXt-V2) | 93.42 | **97.18** | 87.89 | **89.17** | **96.44** | **92.82** | **89.08** | **91.44** |

segmentation dataset. Table IX and Fig. 12, respectively, show the comparative experimental results and segmentation results on the LoveDA dataset. Our method not only achieves the highest values in all three evaluation indicators but also significantly outperforms other methods. This can demonstrate that our method has better adaptability to more challenging scenarios. There are more similar ground objects in LoveDA dataset, and they are easily influenced by various environmental factors and exhibit constantly changing appearance characteristics. Therefore, richer global contextual features are needed to distinguish these similarities and changes. Undoubtedly, our proposed method shows strengths in this regard and can to some extent alleviate the phenomenon of incorrect segmentation. In addition, to further evaluate the ability of our proposed algorithm, we also conducted comparative experiments on the UAVid dataset. From the experimental results as shown in Table X, it can be seen that our proposed GASOT-Net achieved the highest values in all indicators.

### *E. Complexity Analysis*

We choose the model's parameters, multiply–accumulate operations (MACs), and frames per second (FPS) as metrics to evaluate its computational complexity. The comparison of model computational complexity is shown in Table XI.

Early CNN-based representative networks, such as U-Net, SegNet, DANet, and DeepLabV3+, despite having relatively lower computational complexity, demonstrated mediocre segmentation performance. HRNet is specifically tailored for processing high-resolution images, making it highly compatible with the unique characteristics of RS images. However, due to the presence of parallel multiscale branches within the

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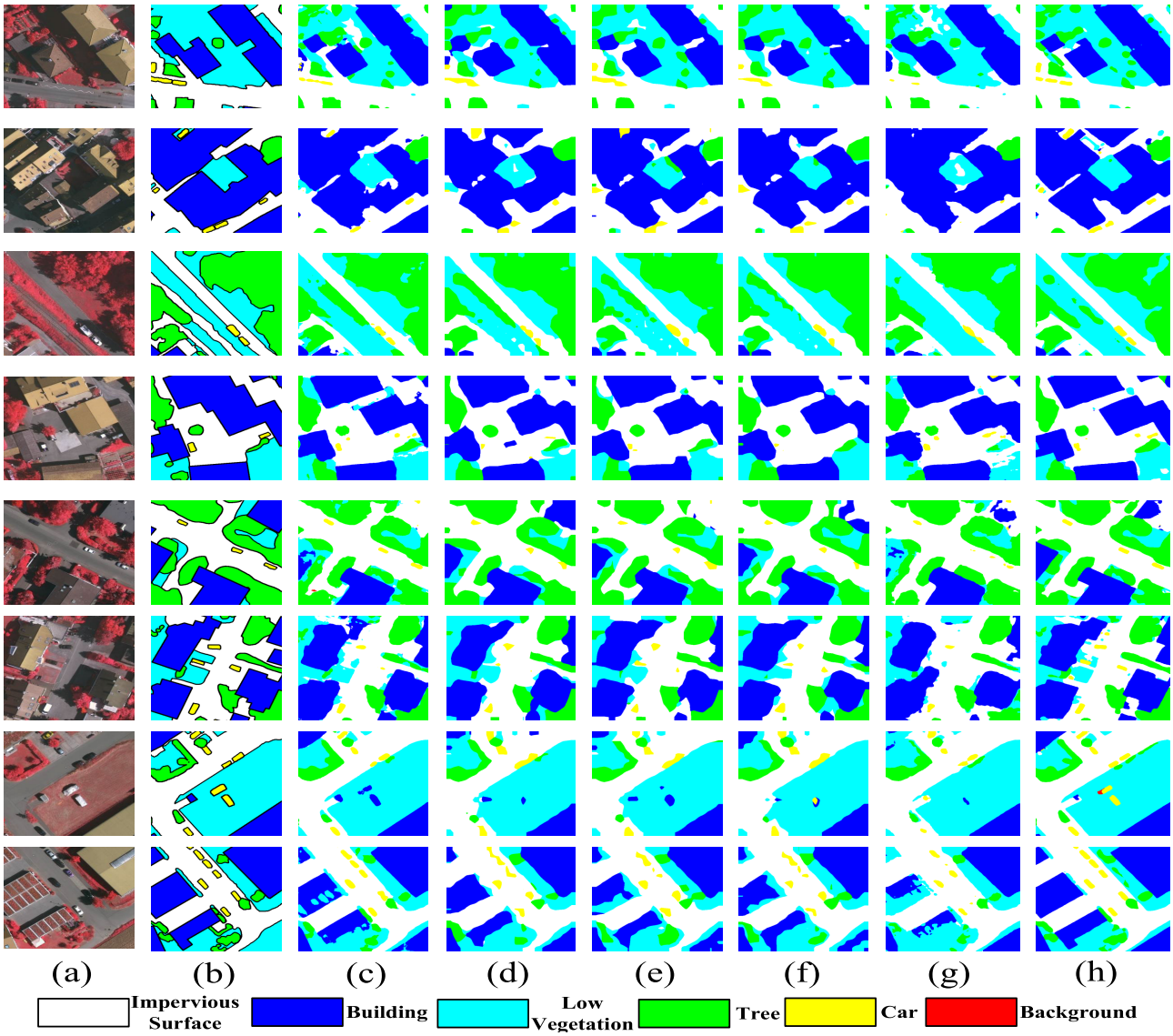



Fig. 10. Comparison of segmentation results on the Vaihingen dataset. (Left–right) (a) Input image. (b) Ground truth. (c) U-Net. (d) DeepLabV3+. (e) DANet. (f) PSPNet. (g) Segformer-B5. (h) GASOT-Net.

TABLE IX
EXPERIMENTAL RESULTS ON THE LOVEDA DATASET

| Method | Background | Building | Road | Water | Barren | Forest | Agriculture | mIoU(%) | mean $F_1$(%) | OA(%) |
|---|---|---|---|---|---|---|---|---|---|---|
| FCN [29] | 50.57 | 44.98 | 47.28 | 64.79 | 33.33 | 37.54 | 55.59 | 47.73 | 64.01 | 67.45 |
| UNet [31] | 43.26 | 37.96 | 43.18 | 48.48 | 7.44 | 36.95 | 46.19 | 37.63 | 53.72 | 59.47 |
| SegNet [30] | 50.81 | 55.04 | 50.65 | 68.79 | 30.42 | 40.54 | 50.60 | 49.55 | 65.52 | 68.02 |
| DeepLabV3+ [33] | 53.58 | 47.15 | 49.34 | 67.22 | 31.63 | 37.24 | 53.34 | 48.50 | 64.61 | 68.40 |
| PSPNet [55] | 53.70 | 49.43 | 48.22 | 69.21 | 34.02 | 38.15 | 56.30 | 49.86 | 65.85 | 69.70 |
| DANet [57] | 52.87 | 52.10 | 50.42 | 70.34 | 30.29 | 40.60 | 52.29 | 49.84 | 65.76 | 69.07 |
| HRNet [36] | 52.04 | 58.33 | 56.01 | 72.33 | 35.34 | 44.16 | 50.63 | 52.69 | 68.37 | 70.09 |
| BiSeNet [79] | 52.02 | 48.15 | 50.24 | 66.01 | 27.79 | 37.57 | 55.22 | 48.14 | 64.16 | 68.00 |
| Segformer-B5 [67] | 43.50 | 39.48 | 38.77 | 42.31 | 7.53 | 33.39 | 42.45 | 35.35 | 50.89 | 57.88 |
| GLOTS [48] | 55.27 | **67.06** | **59.61** | **73.08** | 29.18 | **45.09** | 60.09 | 55.63 | - | - |
| UNetFormer [48] | 42.52 | 60.04 | 50.97 | 55.52 | 6.34 | 35.12 | 46.08 | 42.37 | - | - |
| SwinUNet [48] | 44.64 | 51.73 | 34.92 | 53.97 | 5.43 | 28.73 | 38.15 | 38.37 | - | - |
| Topformer [69] | 43.01 | 39.41 | 38.56 | 41.98 | 7.02 | 32.18 | 42.40 | 34.94 | 50.05 | 57.46 |
| EMRT [1] | 50.58 | 60.48 | 52.78 | 66.73 | 32.88 | 40.28 | 52.51 | 50.89 | 66.78 | 68.27 |
| CM-UNet [85] | 54.56 | 64.13 | 55.51 | 68.06 | 29.62 | 42.92 | 50.42 | 52.17 | - | - |
| RS3Mamba [86] | 39.72 | 58.75 | 57.92 | 61.00 | 37.24 | 39.67 | 33.98 | 50.93 | 66.86 | - |
| GASOT-Net(ConvNeXt-V1) | 55.32 | 64.14 | 58.39 | 72.76 | 33.10 | 41.61 | 62.80 | 55.44 | **70.45** | 73.42 |
| GASOT-Net(ConvNeXt-V2) | **55.71** | 64.22 | 58.64 | 72.89 | **33.26** | 41.84 | **62.98** | **55.65** | 70.41 | **73.45** |

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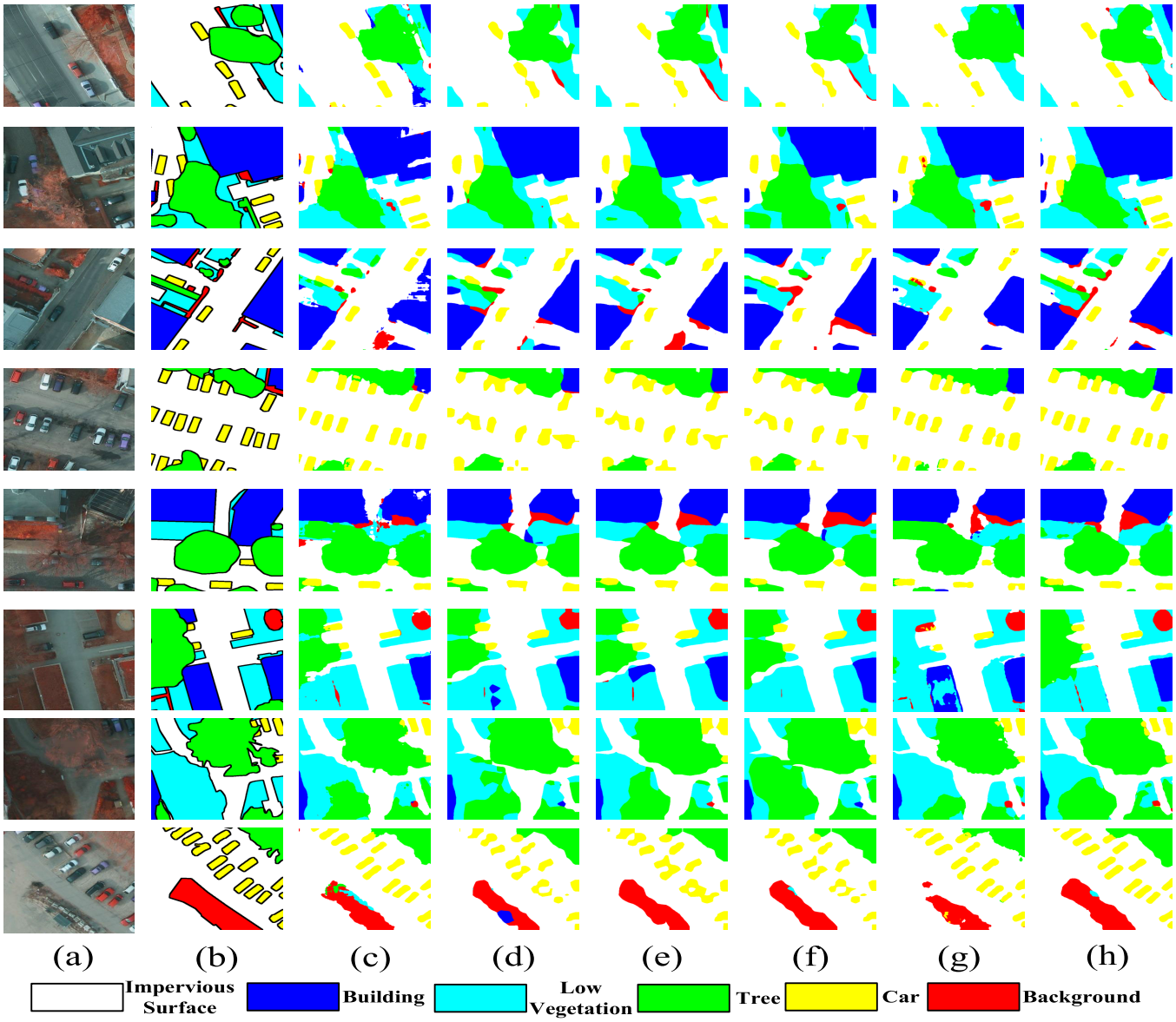



Fig. 11. Comparison of segmentation results on the Potsdam dataset. (Left–right) (a) Input image. (b) Ground truth. (c) U-Net. (d) DeepLabV3+. (e) DANet. (f) PSPNet. (g) Segformer-B5. (h) GASOT-Net.

TABLE X
EXPERIMENTAL RESULTS ON THE UAVID DATASET

| Method | Clutter | Building | Road | Tree | Low Veg. | Mov. Car | Sta. Car | Human | mIoU(%) | OA(%) |
|---|---|---|---|---|---|---|---|---|---|---|
| FCN [29] | 63.91 | 84.72 | 76.51 | 78.32 | 61.88 | 65.87 | 45.54 | 22.26 | 62.38 | 84.55 |
| UNet [31] | 61.80 | 82.94 | 75.15 | 77.27 | 62.03 | 59.59 | 29.98 | 18.62 | 58.42 | 83.43 |
| DeepLabV3+ [33] | 68.86 | 87.62 | 82.22 | 79.76 | 65.88 | 69.86 | 55.39 | 26.07 | 66.96 | 86.94 |
| PSPNet [55] | 68.89 | 87.96 | 82.20 | 81.07 | 65.69 | 70.30 | 56.31 | 24.67 | 67.13 | 87.28 |
| DANet [57] | 64.85 | 85.88 | 77.94 | 78.29 | 61.47 | 59.64 | 47.44 | 9.14 | 60.58 | 84.99 |
| BiSeNetV2 [60] | 61.18 | 81.62 | 77.11 | 75.97 | 61.30 | 66.36 | 38.51 | 15.40 | 59.68 | 83.10 |
| SegNet [30] | 65.62 | 85.89 | 79.23 | 78.78 | 63.73 | 68.94 | 52.10 | 19.29 | 64.20 | 85.54 |
| MFNet [62] | 69.66 | 88.63 | 82.51 | 81.31 | 66.42 | 73.21 | 60.59 | 27.44 | 68.72 | 87.63 |
| Segformer-B5 [67] | 62.10 | 82.76 | 75.10 | 77.13 | 61.98 | 60.10 | 29.55 | 18.43 | 58.39 | 83.16 |
| GASOT-Net(ConvNeXt-V1) | **69.90** | **88.75** | 82.66 | 81.48 | 66.58 | 73.45 | **60.89** | 27.39 | 68.89 | 87.81 |
| GASOT-Net(ConvNeXt-V2) | 69.75 | 88.70 | **82.76** | **81.52** | **66.64** | **73.56** | 60.86 | **27.45** | **68.91** | **87.85** |

network, the FPS is relatively low. In contrast, our network achieves greater accuracy improvements at a comparable FPS level compared with HRNet.

Segformer and TransU-Net are both based on the ViT architecture. Although they exhibit lower parameters and MACs than our method, they do not have a significant advantage in

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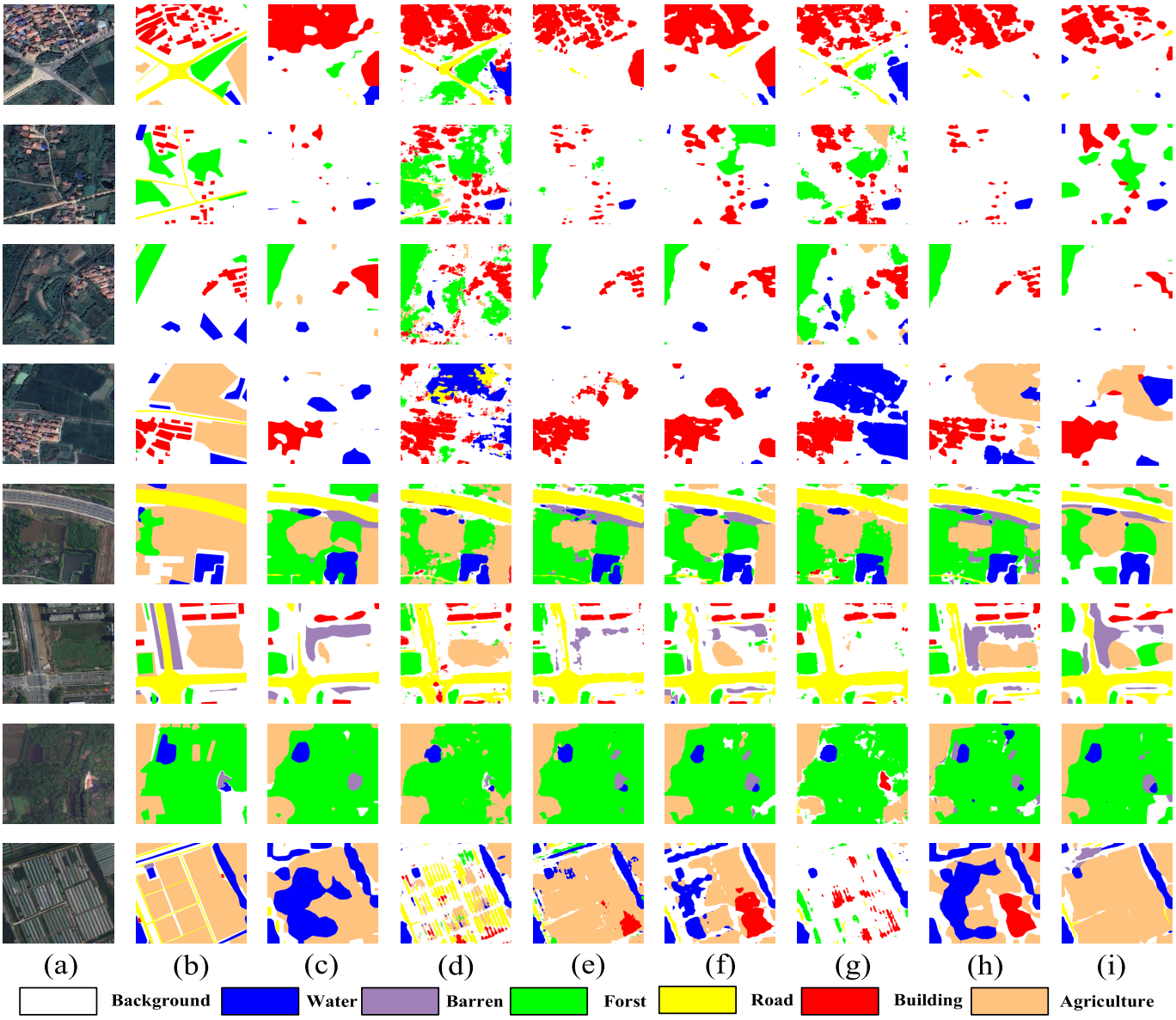



Fig. 12. Comparison of segmentation results on the LoveDA dataset. (Left–right) (a) Input image. (b) Ground truth. (c) FCN. (d) U-Net. (e) DeepLabV3+. (f) DANet. (g) PSPNet. (h) Segformer-B5. (i) GASOT-Net.

TABLE XI
COMPUTATIONAL COMPLEXITY

| Method | Params (M) | Macs (G) | FPS |
|---|---|---|---|
| UNet [31] | 31.04 | 219.13 | 20.08 |
| SegNet [30] | 39.79 | 313.53 | 15.94 |
| DANet [57] | 45.29 | 82.32 | 17.01 |
| DeepLabV3+ [33] | 48.14 | 84.64 | 16.45 |
| HRNet [59] | 65.85 | 93.89 | 9.17 |
| Segformer [68] | 84.6 | 93.89 | 7.31 |
| TransUNet [37] | 105.91 | 148.39 | 12.30 |
| GASOT-Net(ViT) | 166.42 | 266.85 | 12.88 |
| GASOT-Net | 176 | 280.83 | 9.52 |

FPS, and their segmentation performance is much lower than our method.

## V. CONCLUSION

In this article, we propose a GASOT-Net dedicated to mainly deal with the segmentation of highly similar and constantly changing ground objects in RS images. GASOT-Net uses ConvNeXt to extract local features at first, and LFAM is used to aggregate these local features, which can filter out redundant information about local features and highlight key local information. Then, continuous multiple GASOT blocks are used to enrich the representation of global features by adaptively combining first-order information and second-order information. It is worth noting that the GASOT compensates for the shortcoming of ViT in second-order information capture capabilities and can obtain more discriminative information about similar and constantly changing ground objects in RS images. Moreover, GFEM is used to further enhance the representation of global features and achieve precise boundary segmentation of ground objects. Specifically,

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GFEM includes two submodules, FSM and HFFM. FSM can enhance the perception ability of local features by feature offset operations, while HFFM aggregates local and global features from different stages. We conducted extensive ablation and comparative experiments on three famous RS datasets (ISPRS Vaihingen dataset, Potsdam dataset, and LoveDA dataset), and the results show that our proposed GASOT-Net has excellent performance.

In future work, on the one hand, we will continue to delve into the use of transformer structures to mine second-order information in RS images, and on the other hand, we will also strive to improve the efficiency of second-order transformers.

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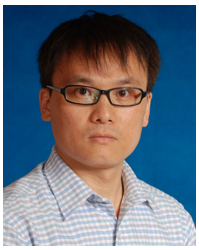
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